📘 SESSION 10 – IMBALANCED DATASET HANDLING

🎯 Topic

- What is Imbalanced Dataset
- Why it is a problem
- Accuracy trap
- Techniques to handle imbalance
- Practical implementation

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1. 🎯 Learning Objectives

By the end of this session, students will be able to:

- Understand what an **imbalanced dataset** is
- Identify problems caused by imbalance
- Understand why **accuracy is misleading**
- Learn techniques like:
  - Oversampling
  - Undersampling
  - Class Weight
- Apply imbalance handling in practical

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2. 🔁 Quick Recap (Connect Previous Session)

Start class like this:

"In the last session, we learned how to evaluate models using accuracy, precision, and recall.
But sometimes accuracy can lie — especially when the dataset is imbalanced.
Today we will understand this problem and how to fix it."

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3. 📖 What is Imbalanced Dataset?

3.1 Definition

An imbalanced dataset is one where **one class has significantly more samples than the other**.

3.2 Simple Meaning

"When one class dominates the dataset, it is called imbalanced data."

Example

Class	Count
Pass	95
Fail	5

👉 Model becomes biased toward majority class

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4. ❗ Why is it a Problem?

Problem 1: Model Bias

Model predicts majority class most of the time

Problem 2: Important Cases Ignored

Minority class often contains:

- Fraud
- Disease
- Failure

👉 Most important cases!

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5. ⚠️ Accuracy Trap (VERY IMPORTANT)

Example

Total Students	100
Pass	95
Fail	5

Model Prediction:

- Predicts ALL as Pass

Accuracy:

$$Accuracy = 95\%$$

👉 Looks good ❌
👉 Actually useless ❌

Teaching Line

"High accuracy does not always mean a good model."

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6. 📊 Metrics That Matter

Instead of accuracy, use:

- Precision
- Recall
- F1 Score

Most Important:

👉 **Recall (for minority class)**

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7. 📊 Techniques to Handle Imbalance

🔹 7.1 Undersampling

Idea

Reduce majority class

Example

Before	After
Pass = 95	Pass = 5
Fail = 5	Fail = 5

Problem

❌ Loss of data

🔹 7.2 Oversampling

Idea

Increase minority class

Example

Before	After
Pass = 95	Pass = 95
Fail = 5	Fail = 95

Advantage

✅ No data loss

Problem

❌ Overfitting risk

🔹 7.3 SMOTE (Concept Only)

Idea

Create synthetic data for minority class

👉 Not required to code deeply now — just concept.

🔹 7.4 Class Weight (Very Important)

Idea

Give more importance to minority class

In model:

LogisticRegression(class_weight='balanced')

Teaching Line

"Instead of changing data, we change importance of classes."

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8. 📊 Real-Life Examples

Fraud Detection

- 99% normal
- 1% fraud

Disease Detection

- Rare disease cases

Spam Detection

- Few spam mails

👉 Always imbalanced problems!

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9. 🎯 Interview Concepts

Q1: What is imbalanced dataset?

👉 One class dominates the dataset.

Q2: Why accuracy fails?

👉 Because model predicts majority class.

Q3: How to handle imbalance?

👉 Oversampling, undersampling, class weight.

Q4: What is SMOTE?

👉 Synthetic data generation technique.

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10. 💻 PRACTICAL (YOUR FORMAT)

🎯 Objective

Handle imbalanced dataset using:

- Normal model
- Balanced model

11. 📊 Output Interpretation

Explain clearly:

Normal Model:

- High accuracy
- Poor minority prediction

Balanced Model:

- Better recall for minority class

👉 Conclusion:

“Balanced model gives more importance to minority class”

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12. ⚠️ Important Teaching Insight

Ask students:

“What is more important — accuracy or catching rare cases?”

👉 Answer:

- Catching rare cases (recall)

“In real-world datasets, imbalance is very common.
A good model is not the one with high accuracy, but the one that correctly identifies important minority cases.”